

The Impact of Interest Rates and Uncertainty on Bank Profits*

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Abstract

While significant literature has been developed on the impact of the level and slope of the yield curve on bank Net-Interest Margin (NIM) and Net Non-Interest Income (NNI), there appears to be a dearth of investigation on the impact of the second moments of economic growth and interest rates on bank profits. We aim to fill that gap in this paper by using within-regressions and a pseudo-DiD approach with US Call Report data to estimate the impact of uncertainty, specifically macroeconomic, and interest-rate uncertainty, on bank profits. We provide evidence for a transmission channel for the impact of macroeconomic uncertainty on the real economy through banks and investigate the impact of uncertainty shocks on financially constrained banks. Our estimates suggest that a 1 standard deviation (sd) macroeconomic uncertainty shock decreases NIM more than twice as much as 1 percentage point (pp) increase in short-term interest rates.

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1 Introduction

Banks are foundational in the modern economy. The stability of the banking system has been an important, and consequently well-covered, problem since the first bank-run model introduced by Diamond and Dybvig (1983). Despite the extensive literature on the banking system, determinants of bank profit variables are still a growing body of work. Bank profits have been identified as a function of several determinants including loan-rate conventions (Hoffmann et al., 2019), market structure, bank risk aversion (Saunders and Schumacher, 2000), monopoly power (Drechsler et al., 2021), average duration of its maturity mismatch (Gomez et al., 2021), and asymmetric rate-adjustments (Gambacorta and Iannotti, 2007). The relationship between monetary policy and bank Net-Interest Margins (NIM) is of particular interest given the importance of identifying potential monetary policy transmission mechanisms through which short-term frictions in the real economy can be smoothed. There are still significant disagreements within the literature regarding the form this relationship takes and the question remains unsettled. An overview of the disagreement can be found in Paul (2023), Drechsler et al. (2021), and English et al. (2012).

While the relationship between interest rates and bank profits has been well-studied, even if inconclusively, we were hardpressed to find studies on the impact of the second moments of macroeconomic growth or interest rates on bank profits. Given financial market participants are necessarily forward-looking and place significant bets on uncertain outcomes in the future, the impact of the level of uncertainty on banks is a crucial gap to fill. In this paper, we contribute to this gap by utilizing US Call Report data and the macroeconomic uncertainty indicator developed by Jurado et al. (2015) to investigate the relationship between uncertainty, specifically interest rate and macroeconomic uncertainty, and bank profits. In addition, we provide updated estimates of the relationship between short-term rates, expected slope of the yield curve, and the term premium, which Paul (2023) argues is crucial to determine the true impact of interest rates.

The variation of interest-rate and macroeconomic uncertainty over time can be observed in Figure 1. Both series appear quite correlated. Interest-rate uncertainty, as measured by the scaled standard deviation of short-term interest rate each quarter, appears to be more volatile than macroeconomic uncertainty, likely since second moments of the short-term yield incorporate information about both financial market frictions and monetary policy changes. While the peaks

of both indicators coincide, interest-rate uncertainty appears to lead during the 2008 Financial Crisis and in the recent pandemic which may indicate an influence on broader macroeconomic uncertainty. Their differences are also pertinent and informative. Interest-rate uncertainty was heightened during the post-dot-com era in the early 2000s when the Fed started cutting rates while macroeconomic uncertainty was relatively subdued. In contrast, macroeconomic uncertainty has remained high since the start of the COVID pandemic whereas interest-rate uncertainty peaked, subsided, and rose again with the Fed policy changes. Figure 2 visualizes the relationship between NIM, Net Non-Interest Income (NNI), and indicators for macroeconomic and interest-rate uncertainty using scatter plots. Between-effects estimators for their simple linear regression are also included. While a comprehensive analysis follows, the simple linear relationships reaffirm one of the main findings of this paper: bank profits are more responsive to macroeconomic uncertainty while being relatively shielded from interest-rate uncertainty.

For our analysis, we use panel data to estimate the response of bank profits to the shape of the yield curve. We also find evidence for a transmission mechanism for macroeconomic uncertainty shocks through banks and to the real economy. In a state of heightened uncertainty, we find evidence for banks reallocating capital from risky loans to risk-free treasuries while accepting lower net interest margins. Finally, we identify financially-constrained banks using proxies and estimate the differential impact of interest rate and uncertainty shocks on their NIM and NNI.

The rest of the paper is organised as follows. The next section discusses the literature on bank determinants, interest rates, uncertainty, and their relationship. Section 3 describes our data and the processing undertaken including cleaning, merger-adjustment, and seasonal-adjustment. Section 4 regresses bank profits on the shape of the yield curve, macroeconomic uncertainty, and interest rate volatility, while controlling for bank and macroeconomic variables. Section 5 discusses transmission channels for an uncertainty shock and provides evidence for a capital reallocation channel. Section 6 considers the effects of uncertainty on financially-constrained institutions and then the final section concludes.

2 Literature Review

The impact of the level and slope of the yield curve on bank profit margins has been extensively covered in the literature with significant divergence in results depending on the adopted

methodology. The divergence can be attributed to different regression specifications, inclusion of different explanatory variables, and datasets. According to the conventional view, banks engage in maturity transformation, leveraging short-term deposits to invest in long-term securities. This view would imply that, on average, the duration of banks' assets is greater than their liabilities, opening them up to interest rate risk. However, Flannery (1981) empirically investigated the impact of interest rate changes on bank profits of 15 American banks and found that their Net Operating Income is insensitive to changes in the level of the prevailing interest rates and interest rate volatility. The findings support the hypothesis that banks effectively hedge their interest rate risk by matching the average duration of their assets to their liabilities. Since then, further studies have corroborated these findings about the insensitivity of bank profits to short-term rates (Drechsler et al., 2021; English, 2002; Flannery, 1983).

The literature explains the conflict between the conventional view and these findings through several channels. Drechsler et al. (2021) argue that banks' market power ensure that bank deposits are stickier than their contractual maturity (Driscoll and Judson, 2013; Flannery and James, 1984; Hannan and Berger, 1991; Hutchison and Pennacchi, 1996; Janosi et al., 1999; Kahn et al., 1999; Neumark and Sharpe, 1992; Yankov, 2014), increasing the average duration of bank liabilities in practice, and also allows banks to offer deposit rates which are lower than the market short-term rate. Purnanandam (2007) shows that the lending volumes in banks which use derivatives to hedge are less exposed to monetary policy shocks. He argues that banks manage their interest rate risk either via duration matching their on-balance sheet assets and liabilities or through the use of interest rate derivatives, and the use of the latter allows them to maintain operational practices which remain unaffected by monetary policy changes. Rampini et al. (2019) find that banks affected by significant shocks to their net worth reduce hedging activities, likely increasing their risk exposures. There is also evidence to show that banks adjust their loan and deposit rates asymmetrically in their favour in response to monetary tightening versus easing (C. E. V. Borio and Fritz, 1995; Gambacorta and Iannotti, 2007; Lim, 2001). Other explanatory channels through which interest rate changes affect NIMs include floating-rate loans (Hoffmann et al., 2019; Ippolito et al., 2018) and the income gap, defined as the difference between the value of assets and liabilities maturing or repricing within a year (Gomez et al., 2021).

As a likely consequence of these confounding variables and the significant heterogeneity among

banks (Drechsler et al., 2021; Hoffmann et al., 2019), several studies have also found a positive relationship between NIMs and the level of interest rates and the yield curve slope. English et al. (2012) also use US Call Report data and find that an increase in the level and slope of the yield curve increases NIMs after controlling for macroeconomic and bank features. They find that banks with a higher share of demand deposits experience a greater increase in NIM if interest rates increase, corroborating the deposit franchise hypothesis. Similarly, C. Borio et al. (2017) use System GMM with a panel dataset including 109 banks across 14 countries and find evidence for a positive and concave relationship between NIM and the level and term structure of interest rates. They argue that the non-linearity in the relationship arises as a result of deposit rates facing a floor at zero and loan volumes being more sensitive to interest rates than deposits. Their arguments in favour of non-linearity in the relationship between bank profits and interest rates informs our regression specification as well.

Finally, there is also evidence to affirm the conventional view that short-term rates and bank profits are inversely related given their maturity transformation function. To demonstrate this, Paul (2023) decomposes the yield on a long-term maturity into the expected path of short-term rates and the term premium¹ and estimates the impulse response of bank profits using level, expected slope and term premium shocks around FOMC meeting days. Post-decomposition, he finds that NIMs respond negatively to both current and expected short-term interest rate shocks and positively to shocks to the term premium. Hence, a positive relationship between NIMs and the observed spread between long-term and short-term rates can be attributed to bank NIMs being more sensitive to an increase in the term premium as opposed to expected spot rates.

In contrast to the impact of the level and slope of the yield curve on bank profitability, literature on the impact of interest rate and macroeconomic uncertainty on bank profits is more difficult to find. Among the few studies which incorporate interest-rate volatility as a regressor (C. Borio et al., 2017; Saunders and Schumacher, 2000), the response of NIM is generally positive. This

¹The term premium TP_t^m for a security of maturity m at time t can be defined as

$$TP_t^m = Y_t^m - \underbrace{E \left[\prod_{k=0}^{m-1} Y_{t+k}^1 \right]}_{EH_t^m}$$

where Y_t^m is the yield on a m period maturity at time t . Since the future path of spot rates is unknown, the term premium acts as a risk compensation for banks engaging in maturity transformation.

suggests that during periods of high interest rate uncertainty, banks price the heightened risk into their loan and deposit rates, likely in an asymmetric fashion. Bretscher et al. (2018) use firm-level and aggregate data to argue that interest-rate uncertainty predicts slowdowns in real activity. Despite hedging opportunities, they depress investments by financially-constrained companies. They argue that firms which rely more on bank and floating rate debt, a group likely including most small companies, reduce investment more due to uncertainty about floating rate payments. Since banks are on the receiving end of such variable rate payments, it seems reasonable to argue that interest rate volatility should positively impact bank NIMs. The relationship between macroeconomic uncertainty and bank profits similarly faces a dearth of investigation, with only a single study directly addressing it as an explanatory variable for bank profits in Turkey using exchange rate volatility as a proxy (Sahin, 2023).

C. Borio et al. (2017) also find a positive impact of interest rate volatility on NNI. NNI includes trading revenues, fees, and commissions. During periods of high interest-rate volatility, banks may be more agile in responding to interest rate conditions and hence perform better at trading and derivative hedging. High levels of uncertainty may also compel other financial firms to seek bank advisory services and products, boosting fees and commissions earned as well. Evidence for a negative relationship between macroeconomic uncertainty and NIMs, on the other hand, is delivered by a recent randomized-control trial conducted by Kumar et al. (2023). They deliver exogenous information treatments (in the form of 1st, 2nd moment shocks and a control) to firms in New Zealand concerning macroeconomic uncertainty and track their survey responses to questions regarding employment, investment, prices, and wages before and six months after receiving the treatment. Their results demonstrate that uncertain firms reduce prices, employment and investment, face decreased sales, and are less likely to open new facilities or invest in new technologies. Quagliariello (2009) also finds that cross-sectional dispersion of bank lending rates decreases during heightened macroeconomic uncertainty. Our results suggest that a similar adjustment process is likely at banks in response to heightened uncertainty, decreasing their loan volumes, and consequently NIMs.

Interest in studying the second moments of macroeconomic variables, specifically economic growth, in the form of macroeconomic uncertainty appears to have been reinvoked by Bloom (2009) who built a model with time-varying uncertainty and used it to estimate the impact of an uncertainty shock on aggregate variables. There have been several macroeconomic uncertainty

indicators developed since. Baker et al. (2016) develop an uncertainty indicator focused on economic policy using news coverage. Rossi and Sekhposyan (2015) use the ex-ante distribution of forecast errors and create uncertainty indices based on the percentiles of the realized errors. In their successive paper, Rossi et al. (2016) use the Survey of Professional Forecasters managed by the Federal Reserve Bank of Philadelphia (2023) to decompose macroeconomic uncertainty into risk and ambiguity which they define as uncertainty about the true probability distribution and support over outcomes. Using impulse responses from a VAR model, they find that an overall increase in uncertainty is recessionary but the impact of its components may differ. The literature has developed several other uncertainty indicators, largely by utilizing factor models. A non-exhaustive list includes Henzel and Rengel (2017), Jo and Sekkel (2019), Popp and Zhang (2016), Cuaresma et al. (2019), and Jurado et al. (2015). A common theme among these indicators is that the uncertainty shocks they identify are far more persistent, in contrast to the theory and empirical findings by Bloom (2009). They also suggest that second moment shocks may be rarer than envisioned, with the Jurado et al. (2015) indicator only identifying four periods with heightened uncertainty post-World War II: months within 1973-74, 1981-1982, the Great Financial Crisis, and the beginning of the Covid Pandemic.

In our analysis, we include the macroeconomic uncertainty indicator developed by Jurado et al. (2015) as a covariate since it is readily available on the authors' website and is based on a large dataset of factor-augmented macroeconomic time-series variables. By including their indicator within our regression specifications, we are able to identify the impact of uncertainty on bank profits, which has not been systematically studied in the literature so far, to the best of our knowledge. Jurado et al. (2015) argue that macroeconomic uncertainty is strongly counter-cyclical and explains a greater proportion of total uncertainty during recessions than non-recessions. A potential concern regarding identification of the impact of uncertainty on bank profits using our methodology may be that it simply acts as a proxy for recessions. However, we control for first moments by including several macroeconomic covariates within our specification which are described in the subsequent section.

3 Data

3.1 Call Reports

The analysis primarily relies on the United States Call Reports which are a compilation of financial forms submitted quarterly by banks and bank-holding companies. The Call Reports form a large panel data with bank-quarter observations. The data starting 1976 to 2010 inclusive is retrieved from the website of the Federal Reserve Bank of Chicago (2011). Data from 2011 to 2023Q2 is retrieved from the Central Data Repository's Public Data Distribution website by the Federal Financial Institutions Examination Council (FFIEC) 2023a.

Details on relevant variable creation, data processing and adjustment can be found in Appendix A and follow English et al. (2012) and Paul (2023). The Python code used to process the data can also be found in our GitHub repository². In summary, the data was processed as follows:

1. The call report data from both sources are combined into a single dataset.
2. All bank-holding companies are removed using bank-level attributes retrieved from FFEIC 2023b.
3. The data is merger-adjusted using the bank transformations recorded by FFEIC 2023b.
4. Income statement variables are seasonally-adjusted using X13-ARIMA.
5. Balance sheet and Income statement items are used to create the relevant variables for our analysis. Income statement (flow) variable observations in Q2-Q4 are differenced at the bank level to convert their reporting frequency from Year-to-Date to Quarterly.
6. The dataset is filtered using the criteria detailed by English et al. (2012) in their Data Appendix and all bank variables are trimmed at the 99% quantile.

At the end of this procedure, we have 427,375 bank-quarter observations from 1997 Q2 to 2023 Q2. A bank is in our sample for an average of 96 quarters. A quarter has 4070 banks on average. Table 1 contains descriptive statistics for our sample after the aforementioned processing.

²<https://github.com/bharatgw/UncertaintyAndBankProfits>

Table 1: Bank Variables Descriptive Statistics

	Mean	P ₀	P ₂₅	P ₅₀	P ₇₅	P ₁₀₀	SD	Skew	Kurt	N
IEA	673962.27	12791.71	71718.5	155523.5	378418	16590277.97	2121668.48	5.99	41.3	427375
Assets	710117.12	13945	77040.5	166569	404776.5	17272675.1	2210260.58	5.97	41.06	427375
Liabilities	632684.18	11824.71	68255.5	148019	361793	15368151.8	1966397.96	5.96	41.02	427375
Equity	69238.38	1214	8089	17493	41882	1611053.33	206988.28	5.91	40.52	427375
Asset Duration ¹	4.55	0.67	2.76	4	5.84	12.46	2.46	1.03	3.82	427375
Liability Duration ¹	0.43	0.03	0.26	0.39	0.55	1.33	0.25	1.09	4.59	427375
Maturity Gap ¹	4.12	0.22	2.32	3.58	5.43	11.88	2.44	0.99	3.7	427375
Loan Share ²	62.79%	23.53%	53.59%	64.28%	73.47%	91.12%	14.57%	-0.46%	2.82%	427375
Demand Deposit Share ³	27.99%	0.2%	17.83%	27.08%	37.16%	65.81%	14.13%	0.35%	2.73%	427375
Saving Deposit Share ³	28.09%	2.01%	17.34%	25.07%	35.12%	78.06%	15.25%	1.11%	4.19%	427375
Time Deposit Share ³	38.8%	3.82%	26.22%	39.04%	50.51%	85.69%	16.96%	0.18%	2.73%	427375
Treasury Share ²	16.06%	0%	6.7%	13.84%	22.9%	53.42%	12.06%	0.91%	3.44%	427375
Other Asset Share ²	15.23%	3.63%	9.39%	12.82%	18.15%	59.99%	9.23%	2.18%	9.46%	427375
Other Liability Share ³	3.2%	0.07%	0.52%	1.16%	3.58%	28.3%	4.91%	2.86%	12.23%	427375
NIM ⁴	0.98%	0.49%	0.85%	0.97%	1.1%	1.74%	0.21%	0.63%	4.31%	427375
NNI ⁴	-0.6%	-1.41%	-0.71%	-0.58%	-0.48%	0.02%	0.21%	-0.65%	5.3%	427375
ROA ⁴	0.26%	-0.73%	0.17%	0.27%	0.37%	0.92%	0.22%	-1.01%	8.16%	427375
Loan Provision ⁴	0.08%	-0.1%	0.01%	0.04%	0.09%	0.96%	0.14%	3.83%	20.57%	266728
$\Delta \ln$ (Assets)	1.39%	-7.77%	-0.41%	1.13%	2.87%	13.17%	3.27%	0.64%	5.18%	427375
IR Hedging Derivatives ⁴	0.54%	0%	0%	0%	0%	17.5%	2.33%	5.71%	37.54%	427371
IR Trading Derivatives ⁴	0.05%	0%	0%	0%	0%	3.33%	0.37%	8.29%	71.9%	427371

All bank variables are trimmed at the 99th percentile over all banks and quarters.

¹ In years

² As a fraction of Total Assets

³ As a fraction of Total Liabilities

⁴ Scaled by lagged Interest-Earning Assets.

3.2 Interest Rates and Macroeconomic Data

The Federal Funds Rate, US Treasury Rates and the Term Premium on Treasury Bonds as estimated by Kim and Wright (2005) are retrieved from the Federal Reserve Economic Data (FRED) website. Motivated by English et al. (2012), we also retrieve several macroeconomic time series which we use as controls from FRED including real GDP, unemployment rate, NAIRU, GDP Price Deflator, CoreLogic House Price Index and the 10-year BBB-Treasury corporate bond credit spread. The S&P500 and VIX were retrieved from WRDS-Compustat.

We measure macroeconomic uncertainty using the indicator developed by Jurado et al. (2015). They define macroeconomic uncertainty as the variance of forecast errors conditional on all current information. It is constructed using a factor-augmented dataset of a large number of macroeconomic variables. Using autoregressive factors and predictors, their forecasting model is equivalent to a Factor-Augmented Vector Autoregression (FAVAR), which they use to estimate the volatility of forecast errors conditional on all information at a given time. The macroeconomic uncertainty time series is retrieved from their website.

The data on implied yield of Federal Fund Futures is retrieved from Bloomberg. The respective Bloomberg Ticker is “FF1.”

4 Macroeconomic Uncertainty and Bank Profits

$$\begin{aligned} \pi_{i,t} = & \rho\pi_{i,t-1} + \beta^T \mathbf{Y}_t^{TP} + \alpha_0 MU_t + \alpha^T \mathbf{Y}_t^{TP} MU_t + \gamma \sigma_t \\ & + \Gamma_0^T \mathbf{B}_{i,t-1} + \Gamma_1^T \mathbf{B}_{i,t-1} Y_t^{3MO} + \Gamma_2^T \mathbf{B}_{i,t-1} (EH_t^{10} - Y_t^{3MO}) + \Theta^T \mathbf{C}_t + \nu_i + u_{i,t} \quad (1) \end{aligned}$$

Equation (1) models Bank Profit variables including NIM, NNI, and Return on Assets (ROA) of bank i in quarter t represented by $\pi_{i,t}$ as a function of the mean level of the yield curve, expected path of future spot rates and the term premium over quarter t , interest rate uncertainty, and macroeconomic uncertainty. We hypothesise that profit variables are persistent and, hence, follow an AR process. The lag order for the dependent variable is selected using AIC and consequently set to 1. The vector of interest rate level and term structure variables is:

$$\mathbf{Y}_t^{TP} = \left[Y_t^{3MO} \quad (EH_t^{10} - Y_t^{3MO}) \quad TP_t^{10} \quad (Y_t^{3MO})^2 \quad (EH_t^{10} - Y_t^{3MO})^2 \quad (TP_t^{10})^2 \right]^T$$

MU_t is the Jurado et al. (2015) Macroeconomic Uncertainty indicator scaled to zero mean and unit standard deviation. Interest-rate uncertainty σ_t is measured by the standard deviation of the short-term rate Y_t^{3MO} over quarter t . We also control for observable bank characteristics in the prior quarter in vector $\mathbf{B}_{t-1}$ ³, their interactions with the level and expected slope of the yield curve, unobserved bank characteristics through fixed-effects ν_i and macroeconomic variables in $\mathbf{C}_t$ described in Section 3.2. Estimation results are presented in table 2.

Our estimation results suggest that short-term interest rates (Y_t^{3MO}) and the expected future path of short-term rates ($EH_t^{10} - Y_t^{3MO}$) are indeed negatively correlated with bank NIMs after controlling for the term premium (TP_t^{10}). These results corroborate the findings within Paul (2023) using bank-level panel data. Our estimates suggest that a 1 pp increase in short-term rates while holding the expected slope of the yield curve constant would result in 9 bp drop in bank profits in the short-term and a 34 bp over the lifetime of the bank. Among the quadratic terms, only $(EH_t^{10} - Y_t^{3MO})^2$ appears to have a statistically significant and negative coefficient, suggesting that expected increases in the slope of the yield curve impact bank NIMs at an increasing rate, potentially because larger movements in the yield curve make it more difficult

³Bank features include Maturity Gap, Other Asset Share, Other Liability Share, Demand Deposit Share, Saving Deposit Share, Loan Share, the log of Total Assets, and a dummy indicating if the pro-forma bank merged with or acquired another bank in the previous quarter.

to immunize bank portfolios by duration-matching.

The coefficient of macroeconomic uncertainty is both statistically and economically significant with an effect size more than twice that of changes in the short-term rate. A 1 standard deviation increase in the macroeconomic uncertainty indicator yields a 70 bp drop in NIMs over the lifetime of the bank. To put the effect size in context, over 1960Q3-2023Q2 the absolute value of the macroeconomic uncertainty indicator crosses the ex-post 1 sd threshold in 57 out of 252 quarters. In addition, changes in the level and expected path of short-term rates also appears to be state-contingent, given the statistically significant interactions between macroeconomic uncertainty and interest-rates. The interacted interest-rate variables take the opposite sign of their individual counterparts, substantiating the countercyclical role of monetary policy. Our results indicate that bank profits increase as a result of contractionary monetary policy during periods of high macroeconomic uncertainty. We argue that that's because heightened macroeconomic uncertainty decreases bank lending which frees up capital for reallocation to treasuries. A contemporaneous increase in the short-rate or expected path of future short-rates boosts the yields on these securities and increases bank NIMs. In the subsequent section, we provide evidence for this adjustment process and substantiate the investments/loan channel as an explanation for the decrease in NIMs due to macroeconomic uncertainty.

It is pertinent that bank profits seem unresponsive to interest rate uncertainty while being negatively impacted by macroeconomic uncertainty. It is likely a result of the fundamental relationship between macroeconomic uncertainty and the broader economy which makes it harder to forecast or hedge. In comparison, interest-rate risk is a well understood phenomenon at banks and, hence, consistently hedged via duration-matching and derivatives. However, further research using better explanatory variables measuring interest rate uncertainty are necessary to make any definitive claims.

5 Uncertainty and Capital Reallocation

We hypothesize that NIMs reduce during heightened periods of macroeconomic uncertainty as lending shrinks and banks shift their capital to lower-risk government securities. This hypothesis is supported by the findings within Kumar et al. (2023) which show that firms reduce investment when faced with an exogenous uncertainty shock. In addition to a reduction in lending, NIMs may also be negatively impacted as default probabilities increase during times of uncertainty,

increasing the provisions banks make for potential loan losses which are incurred as an expense. Realized losses may even be higher than provisioned, decreasing loan interest income further. We investigate these hypotheses using a regression specification similar to 1 with Loan Share, Loan Loss Provisions, Maturity Gap, Loan Income, and Treasury Share as dependent variables. The results are presented in Table 3.

Our results lend partial support to the capital reallocation channel of macroeconomic uncertainty. The impact of macroeconomic uncertainty is only statistically significant on the treasury share of total assets of a bank and positive, in line with our expectations. The impact is economically very significant. 1 sd increase in the uncertainty indicator increases treasury holdings by 4 pp in the same quarter. In contrast, interest rate volatility appears to divert capital away from treasury holdings while increasing loan interest income, suggesting banks likely do price it into their loans. While the coefficients of macroeconomic uncertainty in the other regressions are insignificant, their signs are consistent with our hypothesis except the coefficient of macroeconomic uncertainty in the regression with Loan Interest Income as the dependent variable. It is positive but with a small t-statistic (0.2857). Interestingly, while actual and expected short-rate hikes appear to decrease overall loan share, they appear to increase a bank's maturity gap by 3 quarters on average. The combination of these findings suggests that either banks increase the average duration of the loans they provide following a positive shock to their interest rate expectations, or they purchase other assets, potentially to capitalize on the lower-interest rate environment. Investigating these hypotheses is beyond the scope of this paper.

6 Financially Constrained Banks during Uncertainty

Financially constrained firms have been found to be at a higher risk of default (Purnanandam, 2007), more sensitive to monetary policy changes (Gomez et al., 2021), and disproportionately more exposed to interest-rate risk Bretscher et al. (2018). We predict that the adverse effects of uncertainty on their bottomline is similarly amplified.

We identify financially-constrained banks through two proxies: maturity gap and total assets. Consecutively, we implement a pseudo-Difference-in-Differences estimator as to estimate the effect of interest rate changes and uncertainty shocks on financially constrained banks. To this end, we create dummies for the following three dimensions within our dataset: low maturity gap v. high maturity gap, small v. large bank, and derivative user v. non-derivative user.

We create dummies for the first two by splitting all bank-quarter observations into thirds each quarter using percentiles based on the value of their maturity gap or total assets. Consequently, each group becomes a factor variable. Finally, we discard all observations in the middle group in favour of the two extremes. Similarly, we create a dummy for banks which used derivatives in a given quarter (either for hedging or trading) and banks which did not.

We ran a Wald-test to determine if it is appropriate to include a full-set of dummies (intercept and slope) interacted with all exogenous variables to, in effect, estimate subsample regressions along the factor-dimensions identified. The null hypothesis of common parameters among the subsamples was rejected. Consequently, we estimate the following three regressions for each $g_k \in \{0, 1\}$

$$\begin{aligned}
NIM_{i,t} = & \rho NIM_{i,t-1} + \beta^T \mathbf{Y}_t^{TP} + \alpha_0 MU_t + \alpha^T \mathbf{Y}_t^{TP} MU_t + \gamma \sigma_t \\
& + \Gamma_0^T \mathbf{B}_{i,t-1} + \Gamma_1^T \mathbf{B}_{i,t-1} Y_t^{3MO} + \Gamma_2^T \mathbf{B}_{i,t-1} (EH_t^{10} - Y_t^{3MO}) + \Theta^T \mathbf{C}_t \\
& + g_k \left(\rho_2 NIM_{i,t-1} + \beta_2^T \mathbf{Y}_t^{TP} + \alpha_{0,2} MU_t + \alpha_2^T \mathbf{Y}_t^{TP} MU_t + \gamma_2 \sigma_t \right. \\
& \left. + \Gamma_{0,2}^T \mathbf{B}_{i,t-1} + \Gamma_{1,2}^T \mathbf{B}_{i,t-1} Y_t^{3MO} + \Gamma_{2,2}^T \mathbf{B}_{i,t-1} (EH_t^{10} - Y_t^{3MO}) + \Theta_2^T \mathbf{C}_t \right) + v_i + u_{i,t} \quad (2)
\end{aligned}$$

where $\mathbf{1}(\cdot)$ denotes the indicator function, and

$$\begin{aligned}
g_1 = 1 & \implies (i, t) \in \{(i, t) | \mathbf{1}(\text{MatGap}_{i,t} \geq P_{66}^t)\} \\
g_2 = 1 & \implies (i, t) \in \{(i, t) | \mathbf{1}(\text{AssetSize}_{i,t} \geq P_{66}^t)\} \\
g_3 = 1 & \implies (i, t) \in \{(i, t) | \mathbf{1}(\text{DerivativesNotUsed}_{i,t})\}
\end{aligned}$$

Hence, using the first regression, we can test the statistical significance of the difference between the coefficient estimates of Equation (2) for the bottom and top-third of our sample by Maturity Gap quarterly. The second regression estimates the same but splitting by Asset Size. Finally, the third tests whether the coefficient estimates are different between the subsample of banks which use derivatives in a quarter and the subsample which do not. Estimation results for all three regressions are presented in Table 4.

Larger banks with higher maturity gaps appear to have more persistent NIM series as suggested by the positive and significant coefficients of $g_1 \times L(\text{NIM})$ and $g_2 \times L(\text{NIM})$. Macroeconomic uncertainty shocks have a smaller negative effect on large banks in the top-third of our

dataset relative to the bottom-third by size, as suggested by the significant 17 bp coefficient of $g_1 \times \text{MacroUncertain}$. We argue that that is a likely result within our capital reallocation hypothesis since larger firms are likely to have better capital buffers to continue risky-investment as well as access to suitable investment opportunities. The negative impact of interest rate volatility appears to be similarly mitigated for large banks, though by a significantly smaller magnitude. In contrast, NIM of non-derivative-using banks experiences an additional 3 bp decline relative to banks which use derivatives. Actual and expected short-term rate shocks appear to decrease NIMs more for larger banks. We also include estimation results for NNI as the dependent variable in Equation 2 in Table 5 which suggest that NNI in banks which do not use derivatives to hedge or trade are more responsive to short-rate changes during periods of macroeconomic uncertainty by 6 bp. Surprisingly, NNI in non-derivative using banks is also more persistent than banks which do use derivatives. However, that may be a result of considering banks which do use derivatives to trade in a quarter, hence allowing trading revenues to make NNI more volatile. While the results do not indicate a significant difference between derivative-users and non-derivative-users in terms of the response of their NNI to macroeconomic uncertainty shocks, the coefficient is negative which aligns with the expectation that banks which do not hedge using derivatives would be more sensitive.

7 Conclusion

The impact of slope and level changes in the yield curve on bank profit margins is an open problem in the financial econometrics literature. In this paper, we contribute another determinant for bank profits by analysing the effects of macroeconomic and interest rate uncertainty. We identify and estimate the impact of macroeconomic and interest rate uncertainty on bank profits by using within-effects regressions and a pseudo-Difference-in-Differences approach. First, we estimate the impact of macroeconomic and interest rate uncertainty on the average bank in our sample, and then, on financially-constrained institutions. We find evidence supporting the hypothesis that banks decrease lending in response to an uncertainty shock and reallocate their capital to less risky securities like US treasuries which is in line with adjacent studies like Kumar et al. (2023) and Quagliariello (2009).

Our results also support the intuitive notion that macroeconomic uncertainty has a bigger impact on bank profits relative to interest rate volatility, likely as a consequence of its fundamental

relationship with the broader economy. Our estimates suggest that a 1 sd macroeconomic uncertainty shock reduces bank NIMs by twice as much as 1 pp increase in short-term rate. Finally, we find that larger banks and banks which use derivatives - groups which have significant overlap - appear to be better shielded against macroeconomic and interest rate uncertainty shocks. In contrast, maturity gap does not appear to be a significant predictor of financial constraints in our sample, supplementing Drechsler et al. (2021)'s deposit-franchise hypothesis. However, more refined measures of financial constraints such as income gap and cash flow coverage ratio should be used to more accurately identify financially constrained banks to further the analysis.

References

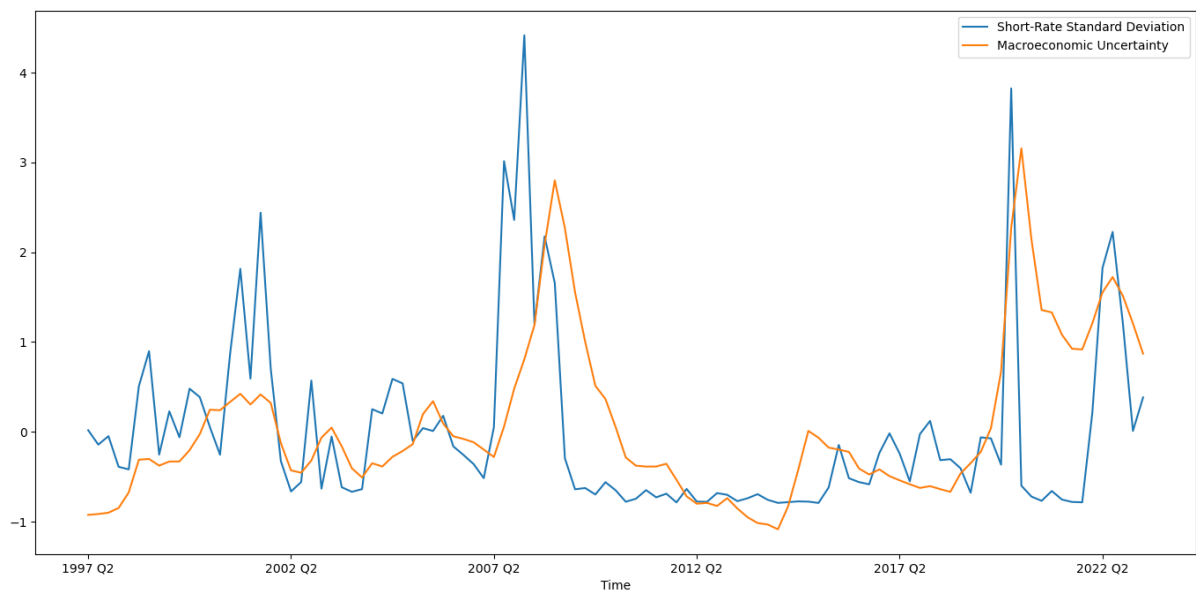
- Baker, S. R., Bloom, N., & Davis, S. J. (2016). Measuring Economic Policy Uncertainty. *The Quarterly Journal of Economics*, 131(4), 1593–1636. <https://doi.org/10.1093/qje/qjw024>
- Bloom, N. (2009). The Impact of Uncertainty Shocks. *Econometrica*, 77(3), 623–685. <https://doi.org/10.3982/ECTA6248>
- Borio, C., Gambacorta, L., & Hofmann, B. (2017). The influence of monetary policy on bank profitability. *International Finance*, 20(1), 48–63. <https://doi.org/10.1111/infi.12104>
- Borio, C. E. V., & Fritz, W. (1995, May). The Response of Short-Term Bank Lending Rates to Policy Rates: A Cross-Country Perspective. <https://doi.org/10.2139/ssrn.863385>
- Bretscher, L., Schmid, L., & Vedolin, A. (2018). Interest Rate Risk Management in Uncertain Times. *The Review of Financial Studies*, 31(8), 3019–3060. <https://doi.org/10.1093/rfs/hhy039>
- Cuaresma, J. C., Huber, F., & Onorante, L. (2019). The Macroeconomic Effects of International Uncertainty. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.3429628>
- Diamond, D. W., & Dybvig, P. H. (1983). Bank Runs, Deposit Insurance, and Liquidity. *Journal of Political Economy*, 91(3), 401–419. Retrieved April 6, 2024, from <https://ideas.repec.org/a/ucp/jpolec/v91y1983i3p401-19.html>
- Drechsler, I., Savov, A., & Schnabl, P. (2021). Banking on Deposits: Maturity Transformation without Interest Rate Risk. *The Journal of Finance*, 76(3), 1091–1143. <https://doi.org/10.1111/jofi.13013>
- Driscoll, J. C., & Judson, R. (2013, October). Sticky Deposit Rates. <https://doi.org/10.2139/ssrn.2241531>
- English, W. B. (2002). Interest rate risk and bank net interest margins. *BIS Quarterly Review*. Retrieved January 13, 2024, from <https://ideas.repec.org/a/bis/bisqtr/0212g.html>
- English, W. B., Van den Heuvel, S., & Zakrajsek, E. (2012, May). Interest Rate Risk and Bank Equity Valuations. <https://doi.org/10.2139/ssrn.2055292>
- Federal Deposit Insurance Corporation. (2018). Quarterly Banking Profile. 13(1), 31–44. Retrieved January 14, 2024, from <https://www.fdic.gov/analysis/quarterly-banking-profile/index.html>

- Federal Financial Institutions Examination Council. (2023a, October). Download Bulk Data - FFIEC Central Data Repository's Public Data Distribution. Retrieved January 14, 2024, from <https://cdr.ffiec.gov/public/PWS/DownloadBulkData.aspx>
- Federal Financial Institutions Examination Council. (2023b, December). Data Download - National Information Center. Retrieved January 14, 2024, from <https://www.ffiec.gov/npw/FinancialReport/DataDownload>
- Federal Reserve Bank of Chicago. (2011, March). Commercial Bank Data - Federal Reserve Bank of Chicago.
- Federal Reserve Bank of Philadelphia. (2023, November). Survey of Professional Forecasters.
- Flannery, M. J. (1981). Market Interest Rates and Commercial Bank Profitability: An Empirical Investigation. *The Journal of Finance*, 36(5), 1085–1101. <https://doi.org/10.1111/j.1540-6261.1981.tb01078.x>
- Flannery, M. J. (1983). Interest Rates and Bank Profitability: Additional Evidence: Note. *Journal of Money, Credit and Banking*, 15(3), 355–362. <https://doi.org/10.2307/1992486>
- Flannery, M. J., & James, C. M. (1984). Market Evidence on the Effective Maturity of Bank Assets and Liabilities. *Journal of Money, Credit and Banking*, 16(4), 435–445. <https://doi.org/10.2307/1992182>
- Gambacorta, L., & Iannotti, S. (2007). Are there asymmetries in the response of bank interest rates to monetary shocks? *Applied Economics*, 39(19), 2503–2517. Retrieved January 23, 2024, from https://econpapers.repec.org/article/tafapplec/v_3a39_3ay_3a2007_3ai_3a19_3ap_3a2503-2517.htm
- Gomez, M., Landier, A., Sraer, D., & Thesmar, D. (2021). Banks' exposure to interest rate risk and the transmission of monetary policy. *Journal of Monetary Economics*, 117, 543–570. <https://doi.org/10.1016/j.jmoneco.2020.03.011>
- Hannan, T. H., & Berger, A. N. (1991). The Rigidity of Prices: Evidence from the Banking Industry. *The American Economic Review*, 81(4), 938–945. Retrieved January 30, 2024, from <https://www.jstor.org/stable/2006653>
- Henzel, S. R., & Rengel, M. (2017). Dimensions of Macroeconomic Uncertainty: A Common Factor Analysis. *Economic Inquiry*, 55(2), 843–877. <https://doi.org/10.1111/ecin.12422>
- Hoffmann, P., Langfield, S., Pierobon, F., & Vuillemeys, G. (2019). Who Bears Interest Rate Risk? *The Review of Financial Studies*, 32(8), 2921–2954. <https://doi.org/10.1093/rfs/hhy113>

- Hutchison, D. E., & Pennacchi, G. G. (1996). Measuring Rents and Interest Rate Risk in Imperfect Financial Markets: The Case of Retail Bank Deposits. *The Journal of Financial and Quantitative Analysis*, 31(3), 399–417. <https://doi.org/10.2307/2331398>
- Ippolito, F., Ozdagli, A. K., & Perez-Orive, A. (2018). The transmission of monetary policy through bank lending: The floating rate channel. *Journal of Monetary Economics*, 95, 49–71. <https://doi.org/10.1016/j.jmoneco.2018.02.001>
- Janosi, T., Jarrow, R. A., & Zullo, F. (1999). An Empirical Analysis of the Jarrow-van Deventer Model for Valuing Non-Maturity Demand Deposits. *The Journal of Derivatives*, 7(1), 8–31. <https://doi.org/10.3905/jod.1999.319107>
- Jo, S., & Sekkel, R. (2019). Macroeconomic Uncertainty Through the Lens of Professional Forecasters. *Journal of Business & Economic Statistics*, 37(3), 436–446. <https://doi.org/10.1080/07350015.2017.1356729>
- Jurado, K., Ludvigson, S. C., & Ng, S. (2015). Measuring Uncertainty. *American Economic Review*, 105(3), 1177–1216. <https://doi.org/10.1257/aer.20131193>
- Kahn, C., Pennacchi, G., & Sopranzetti, B. (1999). Bank Deposit Rate Clustering: Theory and Empirical Evidence. *The Journal of Finance*, 54(6), 2185–2214. <https://doi.org/10.1111/0022-1082.00185>
- Kim, D. H., & Wright, J. H. (2005). An Arbitrage-Free Three-Factor Term Structure Model and the Recent Behavior of Long-Term Yields and Distant-Horizon Forward Rates. Retrieved April 4, 2024, from <https://www.federalreserve.gov/econres/feds/an-arbitrage-free-three-factor-term-structure-model-and-the-recent-behavior-of-long-term-yields-and-distant-horizon-forward.htm>
- Kumar, S., Gorodnichenko, Y., & Coibion, O. (2023). The Effect of Macroeconomic Uncertainty on Firm Decisions. *Econometrica*, 91(4), 1297–1332. <https://doi.org/10.3982/ECTA21004>
- Lim, G. (2001). Bank Interest Rate Adjustments: Are They Asymmetric? *Economic Record*, 77(237), 135–147. <https://doi.org/10.1111/1475-4932.00009>
- Neumark, D., & Sharpe, S. A. (1992). Market Structure and the Nature of Price Rigidity: Evidence from the Market for Consumer Deposits*. *The Quarterly Journal of Economics*, 107(2), 657–680. <https://doi.org/10.2307/2118485>

- Paul, P. (2023). Banks, maturity transformation, and monetary policy. *Journal of Financial Intermediation*, 53(100). Retrieved April 4, 2024, from <https://ideas.repec.org/a/eee/jfinin/v53y2023ics104295732200064x.html>
- Popp, A., & Zhang, F. (2016). The macroeconomic effects of uncertainty shocks: The role of the financial channel. *Journal of Economic Dynamics and Control*, 69, 319–349. <https://doi.org/10.1016/j.jedc.2016.05.021>
- Purnanandam, A. (2007). Interest rate derivatives at commercial banks: An empirical investigation. *Journal of Monetary Economics*, 54(6), 1769–1808. <https://doi.org/10.1016/j.jmoneco.2006.07.009>
- Quagliariello, M. (2009). Macroeconomic uncertainty and banks' lending decisions: The case of Italy. *Applied Economics*, 41(3), 323–336. <https://doi.org/10.1080/00036840601007286>
- Rampini, A. A., Viswanathan, S., & Vuillemy, G. (2019, July). Risk Management in Financial Institutions. <https://doi.org/10.2139/ssrn.2677051>
- Rossi, B., & Sekhposyan, T. (2015). Macroeconomic Uncertainty Indices Based on Nowcast and Forecast Error Distributions. *American Economic Review*, 105(5), 650–655. <https://doi.org/10.1257/aer.p20151124>
- Rossi, B., Sekhposyan, T., & Soupre, M. (2016, July). Understanding the Sources of Macroeconomic Uncertainty. Retrieved March 29, 2024, from <https://papers.ssrn.com/abstract=2816841>
- Sahin, B. C. (2023). Bank Interest Rate Margins and Uncertainty: A Forward-Looking Approach. *Maliye ve Finans Yazıları*, (120), 1–10. <https://doi.org/10.33203/mfy.1105753>
- Saunders, A., & Schumacher, L. (2000). The determinants of bank interest rate margins: An international study. *Journal of International Money and Finance*, 19(6), 813–832. [https://doi.org/10.1016/S0261-5606\(00\)00033-4](https://doi.org/10.1016/S0261-5606(00)00033-4)
- Yankov, V. (2014, December). In Search of a Risk-Free Asset. <https://doi.org/10.2139/ssrn.2544640>

Figure 1: Macroeconomic and Interest Rate Uncertainty over Time



Both indicators are scaled to 0 mean and unit standard deviation.

Table 2: Bank Profits Regressions

π	NIM (1)	NNI (2)	ROA (3)
$L(\pi)$	0.7266*** (0.0171)	0.5295*** (0.0163)	0.4387*** (0.0142)
Y3MO	-0.0009*** (0.0003)	0.0011*** (0.0003)	0.0015** (0.0007)
EH10-Y3MO	-0.0013** (0.0006)	0.0020*** (0.0006)	0.0034*** (0.0013)
TP10	0.0006* (0.0003)	-0.0006* (0.0003)	-0.0014* (0.0007)
Y3MO squared	4.48×10^{-6} (7.34×10^{-6})	$1.6 \times 10^{-5*}$ (8.81×10^{-6})	$2.43 \times 10^{-5**}$ (9.96×10^{-6})
(EH10-Y3MO) square	$-5.68 \times 10^{-5**}$ (2.35×10^{-5})	$-5.15 \times 10^{-5*}$ (2.61×10^{-5})	-0.0001*** (3.62×10^{-5})
TP10 square	-2.29×10^{-5} (4.91×10^{-5})	4.07×10^{-5} (3.74×10^{-5})	9.5×10^{-5} (7.52×10^{-5})
IRVol	8.98×10^{-5} (0.0001)	-0.0002 (0.0001)	-0.0003 (0.0002)
MacroUncertain	-0.0019** (0.0008)	-0.0003 (0.0009)	-0.0019 (0.0019)
Y3MO $\times$ MacroUncertain	0.0005** (0.0002)	3.59×10^{-5} (0.0003)	0.0003 (0.0005)
EH10-Y3MO $\times$ MacroUncertain	0.0011*** (0.0004)	0.0002 (0.0005)	0.0013 (0.0010)
TP10 $\times$ MacroUncertain	-0.0004* (0.0002)	-0.0001 (0.0003)	-0.0005 (0.0005)
Y3MO squared $\times$ MacroUncertain	8.28×10^{-6} (8.32×10^{-6})	$1.62 \times 10^{-5*}$ (9.27×10^{-6})	$5.64 \times 10^{-5***}$ (1.43×10^{-5})
(EH10-Y3MO) square $\times$ MacroUncertain	$-7.73 \times 10^{-5***}$ (2.33×10^{-5})	-1.7×10^{-5} (2.91×10^{-5})	-0.0002*** (3.96×10^{-5})
TP10 square $\times$ MacroUncertain	-0.0003** (0.0001)	-4.96×10^{-5} (5.28×10^{-5})	-0.0003** (0.0001)
Macro Controls	✓	✓	✓
Bank Controls	✓	✓	✓
Observations	420,772	420,772	420,772
R ²	0.8805	0.7269	0.5383
Within R ²	0.7553	0.4151	0.3003
Bank fixed effects	✓	✓	✓

Only selected variable coefficients are reported to conserve space. Driscoll and Kraay standard errors are estimated and reported assuming residuals are autocorrelated up to the 4th lag. $L(\pi)$ denotes the lag of the dependent variable for the respective columns.

Table 3: Explanatory Channel Regressions

	LoanShare (1)	LoanLossProvision (2)	MatGap (3)	IntIncLoans (4)	TreasuryShare (5)
L(LoanShare)	0.9343*** (0.0151)	1.71×10^{-5} (0.0006)	-0.1103 (0.2214)	0.0008 (0.0005)	-0.0104 (0.0119)
Y3MO	-0.0464** (0.0215)	-7.44×10^{-5} (0.0006)	0.2858 (0.2015)	0.0004 (0.0007)	0.0116 (0.0084)
EH10-Y3MO	-0.0864** (0.0398)	-0.0008 (0.0009)	0.7264* (0.3744)	0.0019 (0.0012)	0.0204 (0.0158)
TP10	0.0622*** (0.0230)	-5.52×10^{-6} (0.0005)	-0.2907 (0.1797)	-0.0013* (0.0007)	-0.0066 (0.0097)
Y3MO squared	-0.0005 (0.0005)	-1.95×10^{-5} (1.3×10^{-5})	0.0126*** (0.0044)	5.57×10^{-5} *** (1.25×10^{-5})	6.04×10^{-5} (0.0002)
(EH10-Y3MO) square	-0.0004 (0.0015)	4.5×10^{-5} (3.96×10^{-5})	-0.0387*** (0.0131)	-0.0001*** (3.69×10^{-5})	-0.0001 (0.0006)
TP10 square	-0.0033 (0.0029)	8.69×10^{-5} (6.87×10^{-5})	0.0066 (0.0280)	4.61×10^{-5} (6.68×10^{-5})	0.0016 (0.0012)
IRVol	-0.0090 (0.0090)	0.0002 (0.0003)	0.0288 (0.0745)	0.0004* (0.0002)	-0.0086** (0.0040)
MacroUncertain	-0.0055 (0.0460)	0.0018 (0.0021)	-0.3607 (0.4656)	0.0004 (0.0014)	0.0414** (0.0206)
L(MatGap)	0.0022*** (0.0005)	3.55×10^{-5} (2.67×10^{-5})	0.9370*** (0.0035)	-7.81×10^{-5} *** (1.34×10^{-5})	0.0002 (0.0004)
Y3MO $\times$ MacroUncertain	0.0078 (0.0124)	-0.0004 (0.0006)	0.0978 (0.1225)	-0.0002 (0.0004)	-0.0105* (0.0057)
EH10-Y3MO $\times$ MacroUncertain	-0.0018 (0.0239)	-0.0011 (0.0010)	0.2144 (0.2577)	5.74×10^{-5} (0.0007)	-0.0211** (0.0105)
MacroUncertain $\times$ TP10	-0.0027 (0.0128)	0.0005 (0.0006)	-0.0943 (0.1149)	0.0002 (0.0004)	0.0118** (0.0054)
Y3MO squared $\times$ MacroUncertain	-0.0014*** (0.0005)	-3.48×10^{-5} ** (1.43×10^{-5})	0.0059 (0.0058)	1.46×10^{-5} (1.22×10^{-5})	-0.0003 (0.0003)
(EH10-Y3MO) square $\times$ MacroUncertain	0.0008 (0.0014)	0.0001** (4.38×10^{-5})	-0.0141 (0.0175)	-8.16×10^{-5} ** (3.21×10^{-5})	0.0008 (0.0007)
TP10 square $\times$ MacroUncertain	0.0045 (0.0048)	0.0001 (0.0001)	-0.0470 (0.0633)	-0.0003** (0.0001)	0.0024** (0.0012)
Y3MO $\times$ L(MatGap)	-0.0004*** (0.0001)	-1.5×10^{-5} ** (6.6×10^{-6})		1.4×10^{-6} (3.75×10^{-6})	-2.86×10^{-5} (0.0001)
Y3MO $\times$ L(LoanShare)	0.0042 (0.0038)	0.0003** (0.0001)	-0.0154 (0.0487)	0.0015*** (0.0001)	-7.49×10^{-5} (0.0030)
EH10-Y3MO $\times$ L(MatGap)	-0.0008*** (0.0002)	-2.42×10^{-5} * (1.27×10^{-5})		3.02×10^{-5} *** (6.51×10^{-6})	-2.41×10^{-5} (0.0002)
EH10-Y3MO $\times$ L(LoanShare)	0.0065 (0.0064)	0.0007** (0.0003)	-0.0197 (0.0828)	0.0017*** (0.0002)	-0.0019 (0.0047)
L(LoanLossProvision)		0.3635*** (0.0144)			
L(IntIncLoans)				0.6525*** (0.0200)	
L(TreasuryShare)					0.9475*** (0.0036)
Macro Controls	✓	✓	✓	✓	✓
Bank Controls	✓	✓	✓	✓	✓
Observations	420,772	262,367	420,772	419,732	420,772
R ²	0.9604	0.4316	0.9682	0.9054	0.9708
Within R ²	0.8823	0.2587	0.9124	0.8301	0.9154
Bank fixed effects	✓	✓	✓	✓	✓

Only selected variable coefficients are reported to conserve space. Driscoll and Kraay standard errors are estimated and reported assuming residuals are autocorrelated up to the 4th lag. First lags of all dependent variables are included in their respective regression and reported.

Table 4: Uncertainty Effects on NIM of Financially Constrained Banks

	NIM×MatGap (1)	NIM×AssetSize (2)	NIM×NoDerivative (3)
$g \times L(\text{NIM})$	0.0170** (0.0083)	0.0683*** (0.0157)	0.0127 (0.0110)
$g \times Y3\text{MO}$	0.0004 (0.0003)	-0.0016*** (0.0003)	0.0005 (0.0003)
$g \times \text{EH10-Y3MO}$	0.0008* (0.0005)	-0.0029*** (0.0005)	0.0009 (0.0006)
$g \times \text{TP10}$	-0.0004 (0.0003)	0.0005* (0.0002)	-0.0003 (0.0002)
$g \times Y3\text{MO squared}$	6.76×10^{-6} (7.37×10^{-6})	4.39×10^{-6} (6.32×10^{-6})	3.58×10^{-7} (1.03×10^{-5})
$g \times (\text{EH10-Y3MO}) \text{ square}$	1.06×10^{-6} (2.3×10^{-5})	-1.41×10^{-5} (1.52×10^{-5})	1.1×10^{-5} (2.95×10^{-5})
$g \times \text{TP10 square}$	$7.73 \times 10^{-5***}$ (2.87×10^{-5})	4.45×10^{-5} (3.32×10^{-5})	7.62×10^{-5} (4.99×10^{-5})
$g \times \text{IRVol}$	0.0002 (0.0001)	0.0002** (7.56×10^{-5})	-0.0003** (0.0001)
$g \times \text{MacroUncertain}$	0.0010 (0.0009)	0.0017** (0.0007)	0.0006 (0.0008)
$g \times Y3\text{MO} \times \text{MacroUncertain}$	-0.0004 (0.0002)	-0.0005** (0.0002)	-0.0002 (0.0002)
$g \times \text{EH10-Y3MO} \times \text{MacroUncertain}$	-0.0004 (0.0004)	-0.0009** (0.0004)	-0.0002 (0.0004)
$g \times \text{TP10} \times \text{MacroUncertain}$	0.0003 (0.0002)	0.0005** (0.0002)	0.0002 (0.0002)
$g \times Y3\text{MO squared} \times \text{MacroUncertain}$	1.02×10^{-5} (6.92×10^{-6})	$-1.04 \times 10^{-5**}$ (5.05×10^{-6})	2.23×10^{-5} (1.35×10^{-5})
$g \times (\text{EH10-Y3MO}) \text{ square} \times \text{MacroUncertain}$	$-6.36 \times 10^{-5***}$ (1.85×10^{-5})	2.69×10^{-5} (1.8×10^{-5})	-5.01×10^{-5} (3.47×10^{-5})
$g \times \text{TP10 square} \times \text{MacroUncertain}$	0.0001*** (4.41×10^{-5})	$8.09 \times 10^{-5**}$ (3.38×10^{-5})	0.0001 (6.93×10^{-5})
Macro Controls	✓	✓	✓
Bank Controls	✓	✓	✓
Observations	261,205	276,435	420,772
R ²	0.8903	0.8807	0.8809
Within R ²	0.7506	0.7400	0.7563
Bank fixed effects	✓	✓	✓

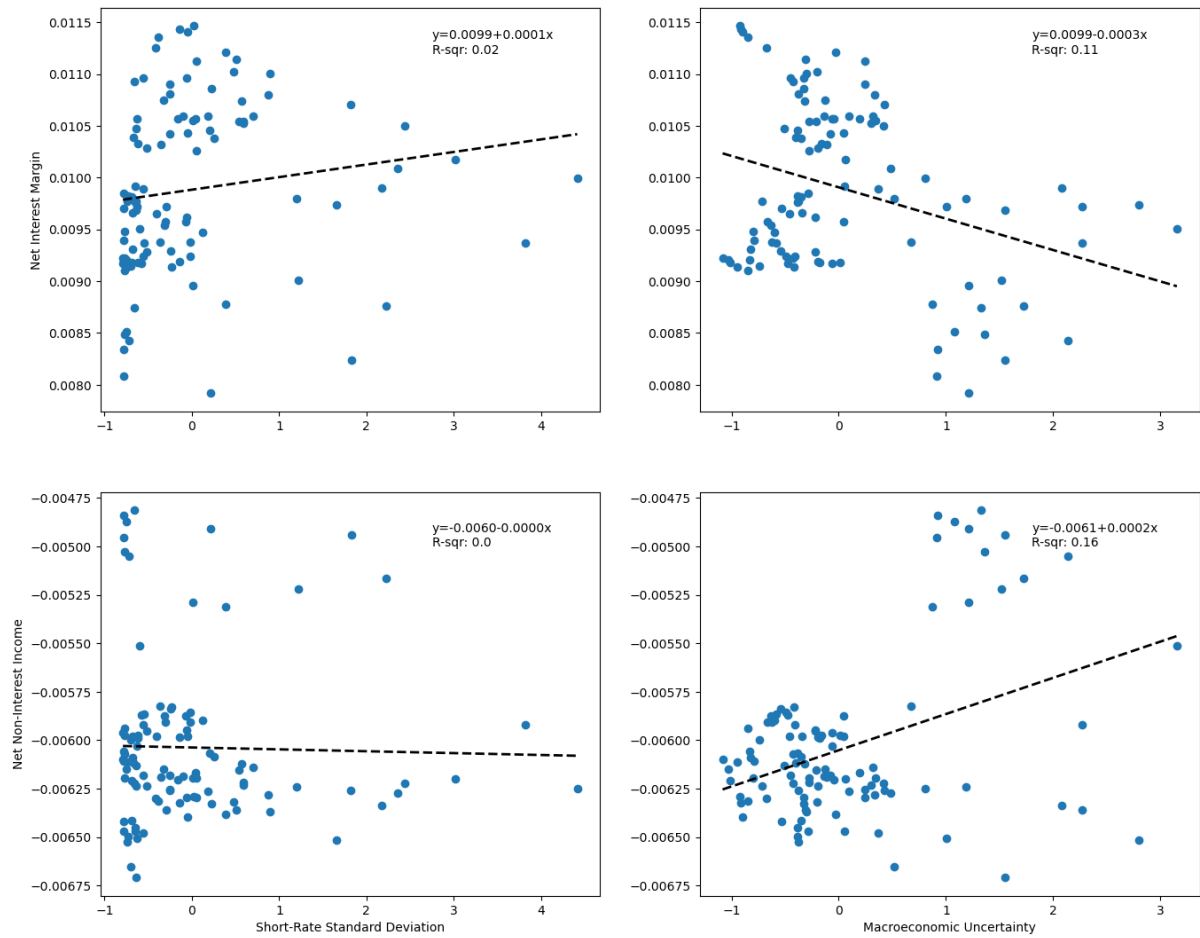
This table presents selected estimates of coefficients from Equation (2), estimated with a full-dummy set (Slope and Intercept) for each exogenous variable. Driscoll and Kraay standard errors are estimated and reported assuming residuals are autocorrelated up to the 4th lag. First lag of the dependent variable - NIM - is included in each regression and reported. g denotes the ‘alternative’ group dummy and its definition varies for each column. For columns (1) and (2), $g=1$ indicates the group of banks each quarter with the highest maturity gap and asset size. For column (3), $g=1$ indicates the group of banks each quarter which do not use derivatives for trading or hedging. Hence, for each regression specification, a significant coefficient suggests that there is difference between sensitivities of the two groups to changes in the given exogenous variable.

Table 5: Uncertainty Effects on NNI of Financially Constrained Banks

	NNI×MatGap (1)	NNI×AssetSize (2)	NNI×NoDerivative (3)
$g \times L(\text{NNI})$	-0.0242** (0.0117)	0.0014 (0.0176)	0.0273** (0.0128)
$g \times Y3\text{MO}$	0.0005 (0.0003)	0.0010*** (0.0003)	0.0006 (0.0004)
$g \times \text{EH10-Y3MO}$	0.0006 (0.0005)	0.0010* (0.0005)	0.0005 (0.0007)
$g \times \text{TP10}$	-0.0003 (0.0002)	0.0002 (0.0002)	-0.0003 (0.0004)
$g \times Y3\text{MO squared}$	-7.79×10^{-7} (3.07×10^{-6})	$-2.24 \times 10^{-5***}$ (5.28×10^{-6})	$-2.71 \times 10^{-5***}$ (6.68×10^{-6})
$g \times (\text{EH10-Y3MO}) \text{ square}$	-1.81×10^{-6} (1.16×10^{-5})	$5.31 \times 10^{-5***}$ (1.49×10^{-5})	$8.6 \times 10^{-5***}$ (2.74×10^{-5})
$g \times \text{TP10 square}$	$5.35 \times 10^{-5***}$ (1.84×10^{-5})	-2.19×10^{-5} (2.71×10^{-5})	-0.0001** (4.3×10^{-5})
$g \times \text{IRVol}$	-2.93×10^{-5} (6.68×10^{-5})	-6.22×10^{-5} (7.1×10^{-5})	7.13×10^{-5} (0.0001)
$g \times \text{MacroUncertain}$	-0.0002 (0.0007)	0.0004 (0.0005)	-0.0014 (0.0011)
$g \times Y3\text{MO} \times \text{MacroUncertain}$	3.91×10^{-5} (0.0002)	-0.0002 (0.0001)	0.0006* (0.0003)
$g \times \text{EH10-Y3MO} \times \text{MacroUncertain}$	7.31×10^{-5} (0.0003)	-6.24×10^{-5} (0.0003)	0.0004 (0.0005)
$g \times \text{TP10} \times \text{MacroUncertain}$	-5.19×10^{-5} (0.0002)	7.79×10^{-5} (0.0002)	-0.0005 (0.0003)
$g \times Y3\text{MO squared} \times \text{MacroUncertain}$	2.68×10^{-6} (3.17×10^{-6})	$2.39 \times 10^{-5***}$ (5.49×10^{-6})	$-6.29 \times 10^{-5***}$ (1.38×10^{-5})
$g \times (\text{EH10-Y3MO}) \text{ square} \times \text{MacroUncertain}$	-5.67×10^{-6} (1.14×10^{-5})	-2.72×10^{-5} (1.93×10^{-5})	0.0001*** (3.45×10^{-5})
$g \times \text{TP10 square} \times \text{MacroUncertain}$	-1.67×10^{-7} (2.95×10^{-5})	-3.66×10^{-5} (3.61×10^{-5})	0.0001* (5.68×10^{-5})
Macro Controls	✓	✓	✓
Bank Controls	✓	✓	✓
Observations	261,205	276,435	420,772
R ²	0.7356	0.7391	0.7277
Within R ²	0.3983	0.3917	0.4167
Bank fixed effects	✓	✓	✓

This table presents selected estimates of coefficients from a regression similar to Equation 2 with NNI as the dependent variable, estimated with a full-dummy set (Slope and Intercept) for each exogenous variable. Driscoll and Kraay standard errors are estimated and reported assuming residuals are auto-correlated up to the 4th lag. First lag of the dependent variable - NNI - is included in each regression and reported. g denotes the ‘alternative’ group dummy and its definition varies for each column. For columns (1) and (2), $g=1$ indicates the group of banks each quarter with the highest maturity gap and asset size. For column (3), $g=1$ indicates the group of banks each quarter which do not use derivatives for trading or hedging. Hence, for each regression specification, a significant coefficient suggests that there is difference between sensitivities of the two groups to changes in the given exogenous variable.

Figure 2: Bank Profits Against Interest-Rate and Macroeconomic Uncertainty



Scatter plots show the time-means of bank profits against scaled macroeconomic uncertainty and interest-rate uncertainty. OLS estimates and R^2 are reported. Bank profits are trimmed at the 99th percentile.

A Data Appendix

We merger-adjust the data for two reasons. Firstly, other papers within the literature most relevant to our work including English et al. (2012) also adjust for mergers, imploring us to do the same to allow valid comparisons between the results. Secondly, merger-adjustment allows us to control for intra-industry asset transfers when analysing the trends concerning the use of the deployed assets to generate revenue and the impact of exogenous variables (such as interest rate changes) on this process. Practically, merger-adjustment removes bank failures in response to interest-rate changes whenever the failing institute is acquired by another bank introducing survivorship bias which is mitigated by retaining institutes which closed without an acquirer. We also include a merger-adjustment dummy for the relevant bank-quarter observations as a bank Acontrol.

While adjusting for mergers in our dataset, we closely follow the Primer released by the Federal Deposit Insurance Corporation (FDIC) 2018. We merger-adjust the dataset through the following steps:

1. We use the transformation record released by FFIEC 2023b to remove company splits and sale of assets (transformation codes 5 and 7) since acquired company continues to exist.
2. All merged and acquired firms within our dataset were mapped to surviving companies as of December 1, 2023.
3. Income statement and balance sheet variables are summed at the acquirer-quarter level.
4. If an institute is acquired by more than one bank then its balance sheet and income statement is assumed to be split evenly between the acquiring institutes and hence divided by the number of acquirers prior to aggregation.

Merger-adjustment reduces the number of banks across our entire dataset from 24,038 to 6,475. Similar to English et al. (2012), the created variables were first seasonally-adjusted and then filtered as follows:

1. All bank-quarter observations with positive asset-growth greater than 20% were eliminated to ensure bank acquisitions from outside the banking industry do not affect the results. It also mitigates the survivorship bias described above.
2. All-banks with an average loan share as a percentage of assets less than 25% were

Table 6: Off-Balance Sheet Variables from Call Reports

Variable Name	Variable Code	Variable Label	Availability
Interest Rate Hedging Derivatives	= RCON8725	Total Gross Notional Amount Of Interest Rate Derivative Contracts Held For Purposes Other Than Trading: Contracts Marked To Market	since 1995:Q2
Interest Rate Trading Derivatives	= RCONA126	Total Gross Notional Amount Of Interest Rate Derivative Contracts Held For Trading	since 1995:Q2

eliminated since these banks do not primarily engage in lending activities.

3. All bank-level variables were trimmed at the 99th percentile over our entire sample to eliminate the influence of outliers or data errors.
4. All banks with less than 24 consecutive quarters of data were eliminated.

Tables 6, 7 and 8 describe the variables used in the analysis and are almost completely reproduced from the Data Appendix within Paul (2023).

Table 7: Balance Sheet Variables from Call Reports

Variable Name	Variable Code	Variable Label	Availability
Total Assets	= RCON2170	Total Assets	since 1976:Q1
Total Liabilities	= RCON2948	Total Liabilities	since 1984:Q1
Total Equity	= RCON3210	Total Equity	since 1976:Q1
Total Loans	= RCON2122	Total Loans	since 1976:Q1
Interest Earning Assets	= RCON0071	Interest-Bearing Balances	since 1976:Q1
	+ RCON2122	Loans and Leases Net of Unearned Income	since 1976:Q1
	+ RCON0390	Investment Securities	1976:Q1-1993:Q4
	+ RCON2146	Assets in Trading Account	1984:Q1-1993:Q4
	+ RCON3545	Trading Assets	since 1994:Q1
	+ RCON1754	Held-to-maturity securities	since 1994:Q1
	+ RCON1772	Available for sale securities	since 1994:Q1
	+ RCON1350	Federal funds sold and securities purchased to re-sell	1976:Q1-2001:Q4
	+ RCONB987	Federal funds sold	since 2002:Q1
	+ RCONB989	Securities purchased to resell	since 2002:Q1
Liquid Assets	= RCON1350	Federal funds sold and securities purchased to re-sell	1976:Q1-2001:Q4
	+ RCONB987	Federal funds sold	since 2002:Q1
	+ RCONB989	Securities purchased to resell	since 2002:Q1
	+ RCON2146	Assets in Trading Account	1984:Q1-1993:Q4
	+ RCON3545	Trading Assets	since 1994:Q1
	+ RCON0390	Investment Securities	1976:Q1-1993:Q4
	+ RCON1754	Held-to-maturity securities	since 1994:Q1
	+ RCON1772	Available for sale securities	since 1994:Q1
Savings Deposits	= RCON6810	Money Market Deposit Accounts	since 1984:Q1
	+ RCON0352	Other Non-transaction Savings Deposits	since 1986:Q1
Demand Deposits	= RCON2215	Total Transaction Accounts	since 1984:Q1
Time Deposits	= RCON6648	Total Time Deposits Of Less Than \$100,000	since 1984:Q1
	+ RCON6645	Time Certificates Of Deposit Of \$100,000 Or More In Domestic Offices	1976:Q1-1997:Q1
	+ RCON6646	Other Time Deposits Of \$100,000 Or More In Domestic Offices	1976:Q1-1996:Q4
	+ RCON2604	Total Time Deposits Of \$100,000 Or More	since 1976:Q1
	+ RCONJ473	Total Time Deposits Of \$100,000 Through \$250,000	since 2010:Q1
	+ RCONJ474	Total Time Deposits Of More Than \$250,000	since 2010:Q1
Interest Bearing Deposits	= RCON6636	Total Interest-Bearing Deposits In Foreign And Domestic Offices	since 1978:Q4
Treasury Securities	= RCONA549	Domestic Government Debt Securities With 3 Months of Remaining Maturity	since 1997:Q2
	+ RCONA550	Domestic Government Debt Securities With 3-12 Months of Remaining Maturity	since 1997:Q2
	+ RCONA551	Domestic Government Debt Securities With 12-36 Months of Remaining Maturity	since 1997:Q2

Table 7 – continued from previous page

Variable Name	Variable Code	Variable Label	Availability
	+ RCONA552	Domestic Government Debt Securities With 36-60 Months of Remaining Maturity	since 1997:Q2
	+ RCONA553	Domestic Government Debt Securities With 60-180 Months of Remaining Maturity	since 1997:Q2
	+ RCONA554	Domestic Government Debt Securities With more than 180 Months of Remaining Maturity	since 1997:Q2
Assets with Maturity Data	= RCONA549	Domestic Government Debt Securities With 3 Months of Remaining Maturity	since 1997:Q2
	+ RCONA550	Domestic Government Debt Securities With 3-12 Months of Remaining Maturity	since 1997:Q2
	+ RCONA551	Domestic Government Debt Securities With 12-36 Months of Remaining Maturity	since 1997:Q2
	+ RCONA552	Domestic Government Debt Securities With 36-60 Months of Remaining Maturity	since 1997:Q2
	+ RCONA553	Domestic Government Debt Securities With 60-180 Months of Remaining Maturity	since 1997:Q2
	+ RCONA554	Domestic Government Debt Securities With more than 180 Months of Remaining Maturity	since 1997:Q2
	+ RCONA555	Mortgage Passthrough Securities With 3 Months of Remaining Maturity	since 1997:Q2
	+ RCONA556	Mortgage Passthrough Securities With 3-12 Months of Remaining Maturity	since 1997:Q2
	+ RCONA557	Mortgage Passthrough Securities With 12-36 Months of Remaining Maturity	since 1997:Q2
	+ RCONA558	Mortgage Passthrough Securities With 36-60 Months of Remaining Maturity	since 1997:Q2
	+ RCONA559	Mortgage Passthrough Securities With 60-180 Months of Remaining Maturity	since 1997:Q2
	+ RCONA560	Mortgage Passthrough Securities With more than 180 Months of Remaining Maturity	since 1997:Q2
	+ RCONA561	Other Mortgage-Backed Securities With Less than 36 Months of Remaining Maturity	since 1997:Q2
	+ RCONA562	Other Mortgage-Backed Securities With More than 36 Months of Remaining Maturity	since 1997:Q2
	+ RCONA564	Closed-End Loans With 3 Months of Remaining Maturity	since 1997:Q2
	+ RCONA565	Closed-End Loans With 3-12 Months of Remaining Maturity	since 1997:Q2
	+ RCONA566	Closed-End Loans With 12-36 Months of Remaining Maturity	since 1997:Q2
	+ RCONA567	Closed-End Loans With 36-60 Months of Remaining Maturity	since 1997:Q2
	+ RCONA568	Closed-End Loans With 60-180 Months of Remaining Maturity	since 1997:Q2
	+ RCONA569	Closed-End Loans With more than 180 Months of Remaining Maturity	since 1997:Q2

Table 7 – continued from previous page

Variable Name	Variable Code	Variable Label	Availability
	+ RCONA570	All Loans and Leases Except Closed-End Loans With 3 Months of Remaining Maturity	since 1997:Q2
	+ RCONA571	All Loans and Leases Except Closed-End Loans With 3-12 Months of Remaining Maturity	since 1997:Q2
	+ RCONA572	All Loans and Leases Except Closed-End Loans With 12-36 Months of Remaining Maturity	since 1997:Q2
	+ RCONA573	All Loans and Leases Except Closed-End Loans With 36-60 Months of Remaining Maturity	since 1997:Q2
	+ RCONA574	All Loans and Leases Except Closed-End Loans With 60-180 Months of Remaining Maturity	since 1997:Q2
	+ RCONA575	All Loans and Leases Except Closed-End Loans With more than 180 Months of Remaining Maturity	since 1997:Q2
Liabilities with Maturity Data	+ RCONA579	Time Deposits <\$100,000 With 3 Months of Remaining Maturity	1997:Q2-2016:Q4
	+ RCONA580	Time Deposits <\$100,000 With 3-12 Months of Remaining Maturity	1997:Q2-2016:Q4
	+ RCONA581	Time Deposits <\$100,000 With 12-36 Months of Remaining Maturity	1997:Q2-2016:Q4
	+ RCONA582	Time Deposits <\$100,000 With more than 36 Months of Remaining Maturity	1997:Q2-2016:Q4
	+ RCONA584	Time Deposits >= \$100,000 With 3 Months of Remaining Maturity	1997:Q2-2016:Q4
	+ RCONA585	Time Deposits >= \$100,000 With 3-12 Months of Remaining Maturity	1997:Q2-2016:Q4
	+ RCONA586	Time Deposits >= \$100,000 With 12-36 Months of Remaining Maturity	1997:Q2-2016:Q4
	+ RCONA587	Time Deposits >= \$100,000 With more than 36 Months of Remaining Maturity	1997:Q2-2016:Q4
	+ RCONHK07	Time Deposits <\$250,000 With 3 Months of Remaining Maturity	since 2017:Q1
	+ RCONHK08	Time Deposits <\$250,000 With 3-12 Months of Remaining Maturity	since 2017:Q1
	+ RCONHK09	Time Deposits <\$250,000 With 12-36 Months of Remaining Maturity	since 2017:Q1
	+ RCONHK10	Time Deposits <\$250,000 With more than 36 Months of Remaining Maturity	since 2017:Q1
	+ RCONHK12	Time Deposits >= \$250,000 With 3 Months of Remaining Maturity	since 2017:Q1
	+ RCONHK13	Time Deposits >= \$250,000 With 3-12 Months of Remaining Maturity	since 2017:Q1
	+ RCONHK14	Time Deposits >= \$250,000 With 12-36 Months of Remaining Maturity	since 2017:Q1
	+ RCONHK15	Time Deposits >= \$250,000 With more than 36 Months of Remaining Maturity	since 2017:Q1

Table 7 – continued from previous page

Variable Name	Variable Code	Variable Label	Availability
	+ RCONF055	FHLB Advances With <12 Months of Remaining Maturity	since 2006:Q3
	+ RCONF056	FHLB Advances With 12-36 Months of Remaining Maturity	since 2006:Q3
	+ RCONF057	FHLB Advances With 36-60 Months of Remaining Maturity	since 2006:Q3
	+ RCONF058	FHLB Advances With more than 60 Months of Remaining Maturity	since 2006:Q3
	+ RCONF060	Other Borrowings With <12 Months of Remaining Maturity	since 2006:Q3
	+ RCONF061	Other Borrowings With 12-36 Months of Remaining Maturity	since 2006:Q3
	+ RCONF062	Other Borrowings With 36-60 Months of Remaining Maturity	since 2006:Q3
	+ RCONF063	Other Borrowings With more than 60 Months of Remaining Maturity	since 2006:Q3
	+ Savings Deposits		
	+ Demand Deposits		
Other Assets	= Total Assets		
	- Assets with Maturity Data		
Other Liabilities	= Total Liabilities		
	- Liabilities with Maturity Data		

Table 8: Income Statement Variables from Call Reports

Variable Name	Variable Code	Variable Label	Availability
Interest Expense (Domestic)	= RIAD4073	Total Interest Expense	since 1984:Q1
	- RIADC900	Total Interest Expense in Foreign Offices	since 2006:Q2
Interest Income (Domestic)	= RIAD4107	Total Interest Income	since 1984:Q1
	- RIADC899	Total Interest Income in Foreign Offices	since 2006:Q2
	- RIADB525	Net Interest Income from International Operations	2001:Q1-2005:Q4
	- RIAD4842	Net Booking Location Adjustment	1984:Q1-2000:Q4
Net Interest Income (Domestic)	= RIAD4074	Net Interest Income	since 1984:Q1
	- RIADC899	Total Interest Income in Foreign Offices	since 2006:Q2
	+ RIADC900	Total Interest Expense in Foreign Offices	since 2006:Q2
	- RIADB525	Net Interest Income from International Operations	2001:Q1-2005:Q4
	- RIAD4842	Net Booking Location Adjustment	1984:Q1-2000:Q4
Net Noninterest Income (Domestic)	= RIAD4079	Total Noninterest Income	since 1984:Q1
	- RIAD4093	Total Noninterest Expense	since 1984:Q1
	- RIADC902	Noninterest Income from Foreign Offices (Trading)	since 2006:Q1
	- RIADC903	Noninterest Income from Foreign Offices (Advisory)	since 2006:Q1
	- RIADC904	Noninterest Income from Foreign Offices (Securitization)	since 2006:Q1
	- RIADC905	Noninterest Income from Foreign Offices (Other)	since 2006:Q1
	+ RIADC907	Total Noninterest Expense in Foreign Offices	since 2006:Q1
	- RIAD4097	Noninterest Income from International Business	1984:Q1-2005:Q4
	+ RIAD4239	Noninterest Expense from International Business	1984:Q1-2005:Q4
Interest Income from Loans	= RIAD4010		since 1976:Q1
Interest Income from Federal Funds Sold and Securities Purchased	= RIAD4020		since 1976:Q1
Interest Income from Other Securities	= RIAD4060		since 1976:Q1
Interest Income from Trading Assets	= RIAD4069		since 1984:Q1
Interest Income on Balances Due from Depository Institutions	= RIAD4115		since 1976:Q1
Interest and Dividend Income from Securities	= RIAD4218		since 1984:Q1
Other Interest Income	= RIAD4518		since 2001:Q1
Interest Expense on Deposits	= RIAD4170		1976:Q1-2016:Q4
Interest Expense on Trading Liabilities	= RIAD4185		since 1978:Q2
Interest Expense on Notes	= RIAD4200		since 1976:Q1
Provision for Bad Loans (Domestic)	= RIAD4230	Total Provision for Loans and Lease Losses	since 1976:Q1

Table 8 – continued from previous page

Variable Name	Variable Code	Variable Label	Availability
	+ RIADJJ33	Total Provision for Loans and Lease Losses	since 2019:Q1
	- RIADKW02	Provision for Loans and Lease Losses in Foreign Offices	since 2019:Q1
	- RIADC901	Provision for Loans and Lease Losses in Foreign Offices	2006:Q1-2018:Q4
	- RIAD4235	Provision for Loans and Lease Losses for International Business	1984:Q1-2005:Q4
Other Interest Expense	= RIADGW44		since 2017:Q1

B Subsample Regressions for Financially Constrained Banks

Here we present the subsample regressions for the subsamples defined in Section 6. The coefficients are easily observed from and comparable across the columns for the respective subsamples. The downside is that we are unable to test the significance of the differences between our coefficients across regressions. For each subsample, we regress NIM and NNI in the same specification as Equation 1. The estimation results are presented in Table 9 for NIM and Table 10 for NNI. As expected, results correspond to our findings in Section 6.

Table 9: NIM Regressions for Financially Constrained Banks

	Full Sample (1)	$P_{33} \geq \text{Maturity Gap}$ (2)	$P_{66} \leq \text{Maturity Gap}$ (3)	$P_{33} \geq \text{Total Assets}$ (4)	$P_{66} \leq \text{Total Assets}$ (5)	IR Derivatives used (6)	IR Derivatives unused (7)
L(NIM)	0.7266*** (0.0171)	0.7000*** (0.0186)	0.7120*** (0.0159)	0.6669*** (0.0222)	0.7339*** (0.0154)	0.6587*** (0.0334)	0.7210*** (0.0165)
Y3MO	-0.0009*** (0.0003)	-0.0011*** (0.0004)	-0.0008*** (0.0003)	0.0001 (0.0004)	-0.0015*** (0.0003)	-0.0013*** (0.0005)	-0.0007*** (0.0003)
EH10-Y3MO	-0.0013** (0.0006)	-0.0017** (0.0007)	-0.0010* (0.0005)	0.0005 (0.0007)	-0.0024*** (0.0006)	-0.0019** (0.0010)	-0.0008 (0.0006)
TP10	0.0006* (0.0003)	0.0008** (0.0004)	0.0005* (0.0003)	0.0003 (0.0004)	0.0008*** (0.0003)	0.0007* (0.0004)	0.0005* (0.0003)
Y3MO squared	4.48×10^{-6} (7.34×10^{-6})	4.07×10^{-6} (9.2×10^{-6})	7.69×10^{-6} (5.83×10^{-6})	-6.62×10^{-8} (8.43×10^{-6})	3.93×10^{-6} (7.24×10^{-6})	6.82×10^{-6} (1.51×10^{-5})	3.22×10^{-6} (6.94×10^{-6})
(EH10-Y3MO) square	-5.68×10^{-5} (2.35×10^{-5})	-6.57×10^{-5} (2.92×10^{-5})	-5.68×10^{-5} (2.04×10^{-5})	-4.36×10^{-5} (2.62×10^{-5})	-5.78×10^{-5} (2.28×10^{-5})	-7.64×10^{-5} (3.92×10^{-5})	-5.27×10^{-5} (2.3×10^{-5})
TP10 square	-2.29×10^{-5} (4.91×10^{-5})	-4.63×10^{-5} (5.03×10^{-5})	2.35×10^{-5} (4.37×10^{-5})	-4.03×10^{-5} (5.48×10^{-5})	4.97×10^{-6} (4.37×10^{-5})	-7.73×10^{-5} (9.47×10^{-5})	-2.06×10^{-5} (4.61×10^{-5})
IRVol	8.98×10^{-5} (0.0001)	-2.84×10^{-5} (0.0002)	0.0001 (0.0001)	-1.77×10^{-5} (0.0001)	0.0002 (0.0001)	0.0003 (0.0002)	4.2×10^{-5} (0.0001)
MacroUncertain	-0.0019** (0.0008)	-0.0022** (0.0011)	-0.0012** (0.0006)	-0.0025*** (0.0008)	-0.0009 (0.0009)	-0.0024** (0.0010)	-0.0018** (0.0008)
Y3MO $\times$ MacroUncertain	0.0005** (0.0002)	0.0007** (0.0003)	0.0003* (0.0002)	0.0007*** (0.0002)	0.0003 (0.0003)	0.0007** (0.0002)	0.0005** (0.0002)
EH10-Y3MO $\times$ MacroUncertain	0.0011*** (0.0004)	0.0011* (0.0006)	0.0008** (0.0003)	0.0014*** (0.0004)	0.0005 (0.0004)	0.0012** (0.0005)	0.0010*** (0.0004)
TP10 $\times$ MacroUncertain	-0.0004* (0.0002)	-0.0005* (0.0003)	-0.0002 (0.0001)	-0.0006** (0.0002)	-0.0001 (0.0002)	-0.0006** (0.0003)	-0.0004* (0.0002)
Y3MO squared $\times$ MacroUncertain	8.28×10^{-6} (8.32×10^{-6})	-2.27×10^{-6} (9.92×10^{-6})	1.05×10^{-5} (7.05×10^{-6})	1.15×10^{-5} (9.34×10^{-6})	9.78×10^{-7} (7.92×10^{-6})	-1.75×10^{-5} (1.49×10^{-5})	7.1×10^{-6} (8×10^{-6})
(EH10-Y3MO) square $\times$ MacroUncertain	-7.73×10^{-5} (2.33×10^{-5})	-2.48×10^{-5} (2.74×10^{-5})	-8.97×10^{-5} (1.98×10^{-5})	-8.16×10^{-5} (2.78×10^{-5})	-5.24×10^{-5} (2.16×10^{-5})	-2.28×10^{-5} (3.83×10^{-5})	-7.39×10^{-5} (2.24×10^{-5})
TP10 square $\times$ MacroUncertain	-0.0003** (0.0001)	-0.0003*** (0.0001)	-0.0002** (8.63×10^{-5})	-0.0003*** (0.0001)	-0.0002** (9.38×10^{-5})	-0.0003** (0.0001)	-0.0003** (9.83×10^{-5})
Macro Controls	✓	✓	✓	✓	✓	✓	✓
Bank Controls	✓	✓	✓	✓	✓	✓	✓
Observations	420,772	139,919	140,332	140,014	140,311	66,798	353,974
R ²	0.8805	0.8729	0.9056	0.8690	0.8929	0.8530	0.8839
Within R ²	0.7553	0.7085	0.7584	0.7255	0.7450	0.6301	0.7528
Bank fixed effects	✓	✓	✓	✓	✓	✓	✓

Only selected variable coefficients are reported to conserve space. Driscoll and Kraay standard errors are estimated and reported assuming residuals are autocorrelated up to the 4th lag.

Table 10: NNI and Macroeconomic Uncertainty Regressions for Financially Constrained Banks

	Full Sample (1)	$P_{33} \geq \text{Maturity Gap}$ (2)	$P_{66} \leq \text{Maturity Gap}$ (3)	$P_{33} \geq \text{Total Assets}$ (4)	$P_{66} \leq \text{Total Assets}$ (5)	IR Derivatives used (6)	IR Derivatives unused (7)
L(NNI)	0.5295*** (0.0163)	0.5082*** (0.0173)	0.4671*** (0.0155)	0.4857*** (0.0130)	0.4963*** (0.0236)	0.4394*** (0.0261)	0.5224*** (0.0140)
Y3MO	0.0011*** (0.0003)	0.0010** (0.0004)	0.0015*** (0.0004)	0.0006* (0.0004)	0.0017*** (0.0004)	0.0007 (0.0006)	0.0010*** (0.0003)
EH10-Y3MO	0.0020*** (0.0006)	0.0019*** (0.0007)	0.0025*** (0.0008)	0.0017** (0.0006)	0.0028*** (0.0007)	0.0017* (0.0010)	0.0018*** (0.0006)
TP10	-0.0006* (0.0003)	-0.0006* (0.0003)	-0.0010** (0.0004)	-0.0008** (0.0004)	-0.0007* (0.0004)	-0.0004 (0.0005)	-0.0007*** (0.0003)
Y3MO squared	1.6×10^{-5} * (8.81×10^{-6})	1.72×10^{-5} * (8.78×10^{-6})	1.81×10^{-5} * (1.03×10^{-5})	2.85×10^{-5} *** (1.08×10^{-5})	7.24×10^{-6} (9.05×10^{-6})	3.38×10^{-5} ** (1.32×10^{-5})	1.37×10^{-5} (9.22×10^{-6})
(EH10-Y3MO) square	-5.15×10^{-5} * (2.61×10^{-5})	-4.74×10^{-5} * (2.55×10^{-5})	-5.66×10^{-5} * (3.21×10^{-5})	-8.58×10^{-5} *** (3.14×10^{-5})	-3.34×10^{-5} (2.72×10^{-5})	-0.0001 ** (4.55×10^{-5})	-4.03×10^{-5} (2.67×10^{-5})
TP10 square	4.07×10^{-5} (3.74×10^{-5})	2.24×10^{-5} (3.84×10^{-5})	7.87×10^{-5} * (4.41×10^{-5})	4.47×10^{-5} (4.73×10^{-5})	3.03×10^{-5} (3.89×10^{-5})	0.0001 *** (4.54×10^{-5})	3.46×10^{-5} (3.86×10^{-5})
IRVol	-0.0002 (0.0001)	-0.0001 (0.0001)	-0.0002 (0.0001)	-0.0001 (0.0001)	-0.0002 (0.0001)	-0.0002 (0.0002)	-0.0001 (9.69×10^{-5})
MacroUncertain	-0.0003 (0.0009)	-7.17×10^{-5} (0.0010)	-0.0005 (0.0010)	-0.0003 (0.0010)	0.0002 (0.0010)	0.0008 (0.0011)	-0.0003 (0.0010)
Y3MO $\times$ MacroUncertain	3.59×10^{-5} (0.0003)	-5.18×10^{-5} (0.0003)	8.37×10^{-5} (0.0003)	6×10^{-5} (0.0003)	-0.0002 (0.0003)	-0.0005 (0.0003)	5.13×10^{-5} (0.0003)
EH10-Y3MO $\times$ MacroUncertain	0.0002 (0.0005)	0.0002 (0.0005)	0.0003 (0.0005)	0.0002 (0.0005)	4.42×10^{-5} (0.0005)	-1.21×10^{-5} (0.0005)	0.0002 (0.0005)
TP10 $\times$ MacroUncertain	-0.0001 (0.0003)	-3.35×10^{-5} (0.0003)	-0.0002 (0.0003)	-9.81×10^{-5} (0.0003)	3.77×10^{-5} (0.0003)	0.0003 (0.0003)	-0.0001 (0.0003)
Y3MO squared $\times$ MacroUncertain	1.62×10^{-5} * (9.27×10^{-6})	1.93×10^{-5} * (9.77×10^{-6})	1.75×10^{-5} * (9.09×10^{-6})	4.85×10^{-6} (9.66×10^{-6})	2.91×10^{-5} *** (1.03×10^{-5})	7.71×10^{-5} *** (1.8×10^{-5})	8.7×10^{-6} (8.91×10^{-6})
(EH10-Y3MO) square $\times$ MacroUncertain	-1.7×10^{-5} (2.91×10^{-5})	-2.88×10^{-5} (2.95×10^{-5})	-2.64×10^{-5} (3.15×10^{-5})	-9.3×10^{-6} (3.01×10^{-5})	-4.16×10^{-5} (3.28×10^{-5})	-0.0001 ** (4.94×10^{-5})	-9.05×10^{-6} (2.79×10^{-5})
TP10 square $\times$ MacroUncertain	-4.96×10^{-5} (5.28×10^{-5})	-4.93×10^{-5} (6.41×10^{-5})	-5.33×10^{-5} (4.42×10^{-5})	-4.19×10^{-5} (6.03×10^{-5})	-7.82×10^{-5} (6.18×10^{-5})	-0.0001 * (8.36×10^{-5})	-2.82×10^{-5} (4.8×10^{-5})
Macro Controls	✓	✓	✓	✓	✓	✓	✓
Bank Controls	✓	✓	✓	✓	✓	✓	✓
Observations	420,772	139,919	140,332	140,014	140,311	66,798	353,974
R ²	0.7269	0.7289	0.7541	0.7193	0.7092	0.6655	0.7347
Within R ²	0.4151	0.3620	0.3642	0.3545	0.3731	0.3289	0.4023
Bank fixed effects	✓	✓	✓	✓	✓	✓	✓

Only selected variable coefficients are reported to conserve space. Driscoll and Kraay standard errors are estimated and reported assuming residuals are autocorrelated up to the 4th lag.